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On the set of non-locally flat points of a submanifold of codimension one*

By ROBION C. KIRBY

THEOREM 1. *Let $f: M^{n-1} \rightarrow N^n$ be an imbedding of a topological $(n-1)$ -manifold M without boundary into a topological n -manifold N without boundary. Let E be the set of points of M at which f is not locally flat. Then, if $n \geq 4$, E cannot be a non-empty subset C^* of a Cantor set where C^* is tame in M and $f(C^*)$ is tame in N .*

Chernavskii [8] and Hutchinson [10] have announced results similar to this, proved using engulfing techniques.

If $\partial M \neq \emptyset$ and $E \subset \partial M$, then the same conclusion holds. Let $\lambda: M \rightarrow [0, 1]$ be a continuous function which is zero on ∂M and non-zero elsewhere. Consider the manifold $\tilde{M}$ in $M \times [-1, 1]$ defined by $\tilde{M} = \{(x, t) \mid -\lambda(x) \leq t \leq \lambda(x)\}$. Since f is flat on the interior of M , f extends to an imbedding $f': \tilde{M} \rightarrow N$ for which $f' \mid \partial \tilde{M}$ is locally flat except at E . Since E is tame in $\partial \tilde{M}$ if it is tame in M , and since $\partial \partial \tilde{M} = \emptyset$, we may apply Theorem 1 to see that E cannot be a subset of a Cantor set.

Suppose $p \in E$ is isolated in E . Choose M' to be a neighborhood of p with $p = E' = E \cap M'$, and let α be a flat arc in M' through p . $f(\alpha - p)$ is flat, so by [7], $f(\alpha)$ is flat, and therefore p and $f(p)$ are tame. Applying Theorem 1 to $f \mid M': M' \rightarrow N$, we see that $p \notin E' \subset E$. Thus E has no isolated points and hence must be uncountable, since a closed countable subset of a manifold must contain isolated points. This fact also follows from the theorem that the union of flat cells is flat in codimension one (see [13] or [14]).

The fact that $f(C^*)$ is tame in N cannot be removed from the hypothesis of Theorem 1. A counter-example in R^n could be constructed from Blankinship's arc [1]. This arc is wild at a Cantor set, but lies in an $(n-1)$ -sphere which is locally flat except at the Cantor set; furthermore, this Cantor set is tame in the $(n-1)$ -sphere.

COROLLARY. *Theorem 1 holds in the special case that $M^{n-1} = S^{n-1}$ and $N^n = R^n$ or S^n .*

THEOREM 2. *The corollary is equivalent to Theorem 1.*

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We will prove Theorem 1 by proving Theorem 2 and proving the corollary independently of Theorem 1.

The corollary is a generalization of the Schoenflies theorem which states: *if $f: S^{n-1} \rightarrow S^n$ is a locally flat imbedding, then $f(S^{n-1})$ bounds two closed n -balls in S^n (see Brown [2] or Mazur [12]).* If $n \geq 4$ and f is assumed to be locally flat except at one point, then again $f(S^{n-1})$ bounds two closed n -balls (see Cantrell [5] and [6]). Thus E cannot consist of one point. If $n = 3$, the boundary of a thickened Fox-Artin arc (see [9]) is a 2-sphere with one non-locally flat point. The corollary then resolves the problem when f is locally flat except at a Cantor set which is tame in S^{n-1} and S^n .

The following theorem is used in proving Theorem 2 and the corollary.

THEOREM 3. *Let $f: B^{n-1} \rightarrow \Sigma^n$ be an imbedding which is locally flat except at $0 \in B^{n-1}$. Let $n \geq 4$. Then f extends to an imbedding of Σ^{n-1} into Σ^n which is locally flat except at 0 and ∞ .*

In § 1 we fix notation and present definitions; § 2 contains the proof of Theorem 2 that the corollary implies Theorem 1; § 3 contains the proof of the corollary; and § 4 the proof of Theorem 3.

The author is indebted to David Gillman and Chris Lacher for many enlightening conversations, and to the referee for a very nice simplification of the author's proof of Theorem 3.

1. In euclidean n -space R^n , let rB^n be the ball of radius r , $\{x \in R^n \mid |x| \leq r\}$, and let $rS^{n-1} = \partial rB^n$. We shall consider R^k , $k \leq n$, to be naturally included in R^n as the subset $\{x \in R^n \mid x_{k+1} = \dots = x_n = 0\}$. This provides inclusions $rB^k \subset rB^n$ and $rS^k \subset rS^n$. Occasionally it will be easier to use the n -sphere Σ^n , obtained by compactifying R^n by adding the point ∞ . Then we have natural inclusions $R^k \subset \Sigma^k \subset \Sigma^n$.

Let $\bar{Y}$ denote the closure of Y , $\text{int } Y$ the interior, and ∂Y the boundary. If $X \subset Y \subset R^n$, let $N_\epsilon(X) = \{y \in Y \mid d(y, X) < \epsilon\}$ where d refers to the metric in R^n .

A subset C^* of a Cantor set in the interior or in the boundary of a topological manifold is said to be tame if it lies in a flat arc in the interior or the boundary respectively. If the manifold is combinatorial, this is equivalent to the existence of a proper homeomorphism of the manifold taking C^* into a polyhedral arc. A set is locally tame if it is tame in a polyhedral neighborhood of each point of the set.

If $g: K^k \rightarrow N^n$ is an imbedding of a k -manifold into an n -manifold, then g is said to be locally flat at $p \in K$ or at $f(p) \in N$ if there exists a neighborhood U of $f(p)$ in N such that $(U, U \cap f(K))$ is pairwise homeomorphic to (R^n, R^k) .

If $\partial K = \emptyset$, then g is flat if g extends to an imbedding of $K \times B^{n-k}$ into N . If $K = B^k$, then g is flat if it extends to an imbedding of $2B^n$ into N .

We shall often make use of the facts that a locally flat imbedding is flat in codimension one [3], a locally flat imbedding of B^k in N^n is flat [11], and an arc in N^n , $n \geq 4$, which is locally flat except at isolated points, is flat [7].

2. Proof of Theorem 2. We need to show that Theorem 1 follows from the corollary.

Since C^* is tame in M , it lies on a flat arc J^* in M . Then J^* lies in the interior of an $(n-1)$ -ball B^* in M . $f(C^*)$ is tame, so it lies on a flat arc J' , which lies on the interior of an n -ball B' in N .

There exists a closed neighborhood U of C^* in J^* such that $f(U) \subset \text{int } B'$. $f(U - C^*)$ is locally flat, hence locally tame. Then since J' is tame in the combinatorial manifold B' , we may apply [4, Th. 1] to conclude that $f(U)$ is tame in B' . Now $f(J^*)$ is locally flat everywhere and therefore flat in N . Thus we may assume that the sets $C^* \subset J^* \subset B^*$ are all imbedded in a coordinate patch of N homeomorphic to R^n and therefore denoted by R^n . Also identify B^* with B^{n-1} , assume that $0 \in J^* \subset B^{n-1}$, and recall that $R^n \subset \Sigma^n$.

Since $f(J^*)$ is flat in Σ^n , we can shrink it to the point $f(0)$ via a map s which is a homeomorphism from $\Sigma^n - f(J^*)$ to $\Sigma^n - f(0)$ and is the identity on a neighborhood of $f(\partial B^*)$. We apply Theorem 3 to extend $sf|B^{n-1}$ to an imbedding $F': \Sigma^{n-1} \rightarrow \Sigma^n$ which is locally flat except at 0 and ∞ . Define $F: \Sigma^{n-1} \rightarrow \Sigma^n$ by piecing together $f|B^{n-1}$ and $s^{-1}F'| \Sigma^{n-1} - 0$, noting that on $B^{n-1} - 0$, $s^{-1}F' = s^{-1}sf = f$. F is locally flat except at $C^* \cup \infty$. But $C^* \cup \infty$ is a subset of a Cantor set and tame in Σ^{n-1} and Σ^n . By the corollary F and hence f are locally flat at C^* .

3. Let C be a subset of the standard Cantor set in $[-1, 1]$ got by deleting middle thirds and suppose $-1 \in C$ and $1 \in C$. Let $D = \{d_1, d_2, \dots\}$ be a countable dense subset in the complement of the Cantor set in $[-1, 1]$. Let $[-1, 1]$ be identified with the subset

$$J = \{(x_1, \dots, x_n) \in S^{n-1} \subset R^n \mid x_{n-1}^2 + x_n^2 = 1, \\ x_{n-1} \geq 0, x_1 = x_2 = \dots = x_{n-2} = 0\},$$

by the identification $t \mapsto (0, 0, \dots, \sqrt{1-t^2}, t)$. Then C can be thought of as a subset of J . Let A be an annulus pinched at C , that is, $A = S^{n-1} \times [-1, 1] / \sim$ where $(x, s) \sim (x, 0)$ if $x \in C$. Let $S_t^{n-2} = \{(x_1, \dots, x_n) \in S^{n-1} \mid x_n = t\}$ and $\Sigma_t = S_t^{n-2} \times 1/2$ in A . Also let $A_{1/2} = \{(x, s) \in A \mid s \leq 1/2\}$. See Figure 1.

If C' is any other subset of a Cantor set, we can construct in the same way a pinched annulus A' .

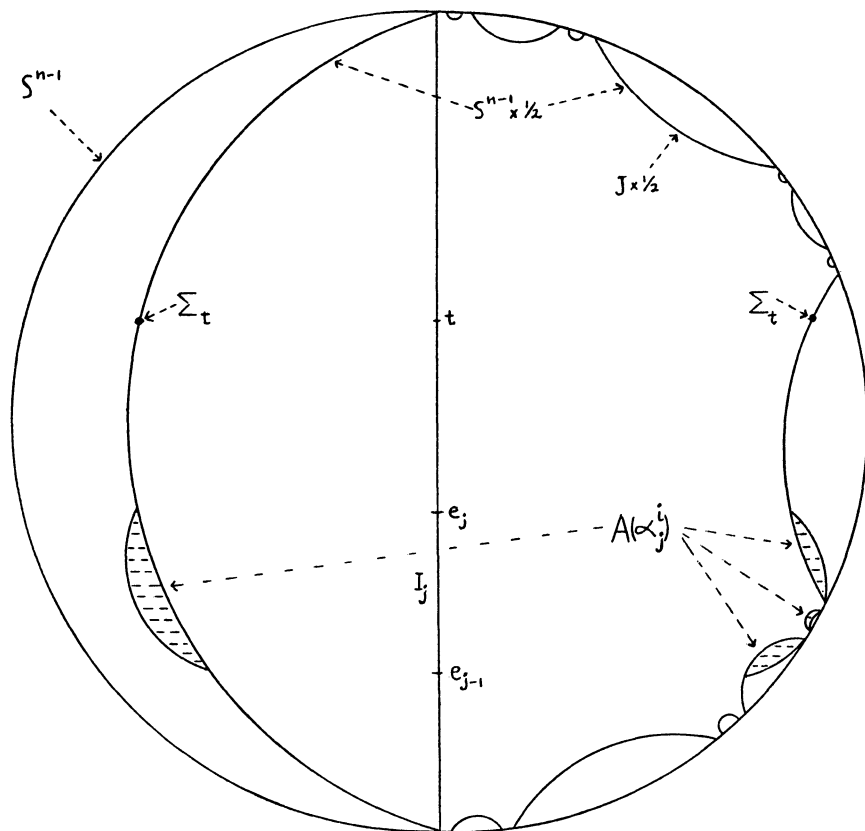


Figure 1

PROOF OF THE COROLLARY. $f(S^{n-1})$ separates R^n into a bounded open set Z “inside” $f(S^{n-1})$ and an unbounded set “outside”. Since C^* is tame in S^{n-1} , there is a homeomorphism of S^{n-1} taking C^* onto C , so we assume $C^* = C$. $f|(S^{n-1} - C)$ is locally flat, so it is flat, so we can assume f imbeds the pinched annulus A into R^n with $f(S^{n-1} \times 1)$ in $\bar{Z}$.

Since $f|(J - C)$ is locally flat and $f(C)$ is tame we can conclude from [4] that $f(J)$ is flat in R^n .

Let $L = f(J \times 1/2)$; it is flat. Denote points of R^n lying on L by t_L , $t \in [-1, 1]$.

The idea of the proof is to find a map $K: R^n \rightarrow R^n$ such that K is a homeomorphism outside $f(S^{n-1} \times 1/2)$, is the identity outside $f(S^{n-1})$, and collapses $f(S^{n-1} \times 1/2)$ to L , taking $f(\Sigma_t)$ to t_L (see (D) below). This “fills up” $\bar{Z}$, which is then shown to be an n -ball (see (E) below). Then f will be locally flat at C , completing the proof of the corollary.

(A) Z is homeomorphic to R^n .

PROOF. Since $f(J)$ is flat, we may shrink it to a point p by a map $g: R^n \rightarrow R^n$ where g is a homeomorphism off $f(J)$. Then $gf(S^{n-1})$ is an $(n-1)$ -sphere which is locally flat except at p . By [5], $gf(S^{n-1})$ bounds an n -ball. In particular, the interior $g(Z)$, is homeomorphic to R^n , so Z is also homeomorphic to R^n .

(B) Given $\varepsilon > 0$, there exists a homeomorphism $G: R^n \rightarrow R^n$ for which

(1) $G = \text{identity outside } f(S^{n-1})$,

(2) $Gf(S^{n-1} \times 1/2) \subset N_\varepsilon(L)$.

Furthermore, this holds for any other C' and A' .

PROOF. Let s be the map which shrinks L to a point p and is a homeomorphism elsewhere. $sf(S^{n-1} \times 1/2)$ is an $(n-1)$ -sphere with only one non-locally flat point p . Then it bounds an n -ball by [6]. So each point $x \in sf(S^{n-1} \times 1/2)$ is joined by a ray r_x to p . Under s^{-1} (not defined at p) these rays are taken to rays called $s^{-1}r_x$, which may not converge to any point of L , but do get arbitrarily close to L . Using these rays, we may slide $f(S^{n-1} \times 1/2)$ into $N_\varepsilon(L)$, fixing L . The result of this slide is called G .

(C) For any $t \in (-1, 1) - C$ and hence for any $t \in D$, there exists a map $H: R^n \rightarrow R^n$ which satisfies

(1) $Hf(\Sigma_t) = t_L$,

(2) $H = \text{identity outside } f(S^{n-1})$,

(3) H is a homeomorphism outside $f(S^{n-1} \times 1/2)$ and on a neighborhood of $f(S^{n-1} \times (1/2) - \Sigma_t)$.

It is implied by (3) that Hf is a locally flat imbedding of $(S^{n-1} \times 1/2) - C - \Sigma_t$. Also, (C) holds for other C' and A' .

PROOF. The set $S_t^{n-2} \times (0, 1/2]$ in A may be identified with $B^{n-1} - 0$. Then $f(B^{n-1} - 0) = f(S_t^{n-2} \times (0, 1/2])$ lies in Z as a closed, flat submanifold. Adding ∞ to $Z = R^n$, we obtain Σ^n . Letting $f(0) = \infty$, f imbeds B^{n-1} and is flat on $B^{n-1} - 0$. By Theorem 3, f extends to an imbedding $f^*: \Sigma^{n-1} \rightarrow \Sigma^n$, flat on $\Sigma^{n-1} - \{0, \infty\}$. We have $f^*(\Sigma^{n-1} - 0) \subset Z$. We can ensure that

$$f^*(\Sigma^{n-1} - 0) \cap f(A_{1/2}) = f^*(B^{n-1} - 0) = f\left(S_t^{n-2} \times \left(0, \frac{1}{2}\right]\right)$$

by pushing $f^*(\Sigma^{n-1} - B^{n-1})$ off $f(A_{1/2})$ using the collaring of $f(S^{n-1})$.

Now let α be the shortest geodesic in Σ^{n-1} joining $(f^*)^{-1}(t_L)$ to ∞ . Then $f^*(\alpha)$ is locally flat mod $f(\infty)$, hence flat. Let $s: Z \rightarrow Z$ be the map which shrinks $f^*(\alpha)$ to t_L , is the identity on $f(A_{1/2})$, and is a homeomorphism off $f^*(\alpha)$. Then $f(S_t^{n-2} \times 1/2) = f(\Sigma_t)$ bounds $sf^*(\Sigma^{n-1} - \text{int } B^{n-1})$, an $(n-1)$ -ball which is flat except at t_L , and whose interior misses $f(A_{1/2})$. It is now easy to construct H by shrinking this $(n-1)$ -ball to t_L .

(D) There exists a continuous map $K: R^n \rightarrow R^n$ which satisfies

- (1) $K = \text{identity outside } f(S^{n-1})$,
- (2) K is a homeomorphism outside $f(S^{n-1} \times 1/2)$,
- (3) $Kf(\Sigma_t) = t_L$ for all $t \in [-1, 1]$.

PROOF. The idea is to deform $f(A_{1/2})$ on to $\bar{Z}$ by collapsing $f(S^{n-1} \times 1/2)$ onto L , taking $f(\Sigma_t)$ to t_L for all t . First, using (B), we move $f(S^{n-1} \times 1/2)$ close to L . Then using (C) we pinch $f(\Sigma_t)$ to t_L for $t = d_1$. This splits the imbedding into two parts, the part between -1 and d_1 , and the part between d_1 and 1 . We proceed in the same way with each part. Thus each Σ_t is pulled closer and closer to t_L and one by one each $f(\Sigma_t)$ for $t \in D$ is taken to t_L . The limit of this process will be K .

Recall that in (B) we were able to move $f(S^{n-1} \times 1/2)$ to L , but in a non-convergent manner; that is, $f(\Sigma_t)$ would not go to t_L , but would oscillate up and down L . We now prevent this by splitting the imbedding into smaller and smaller parts and thereby restricting the motion of $f(\Sigma_t)$ to its appropriate part. This forces $f(\Sigma_t)$ to converge to t_L .

For a given integer $i > 0$, let $e_1, \dots, e_{i-1}$ be a reordering of $d_1, \dots, d_{i-1}$ so that $e_j < e_{j+1}$ for $j = 1, \dots, i-2$. Then $[-1, 1]$ is subdivided into intervals $I_1 = [-1, e_1]$, $I_2 = [e_1, e_2]$, $\dots$, $I_{i-1} = [e_{i-2}, e_{i-1}]$ and $I_i = [e_{i-1}, 1]$. Let $I_{jL} = \{t_L \in L \mid t \in I_j\}$.

Let $\alpha^i: [-1, 1] \rightarrow [0, 1/2]$ be a continuous map for which $\alpha^i(-1) = \alpha^i(1) = \alpha^i(e_j) = 1/2$, for $j = 1, \dots, i-1$, and $\alpha^i(t) \neq 1/2$ otherwise. Also let $\alpha^i(t) \geq \alpha^{i-1}(t)$ for all i and $t \in [-1, 1]$. Let $\alpha_j^i = \alpha^i \mid I_j$. Let

$$A(\alpha_j^i) = \left\{ (x, s) \in A \mid x_n \in I_j, \alpha_j^i(x_n) \leq s \leq \frac{1}{2} \right\}$$

where x_n is the n^{th} coordinate of $x \in S^{n-1}$. Then each $A(\alpha_j^i)$ looks like a cylinder $S^{n-2} \times I$ fattened except at the top and bottom and at points of C (except that $A(\alpha_0^i)$ has its bottom pinched and $A(\alpha_i^i)$ its top pinched to a point). See Figure 1. Furthermore, if the top and bottom are pinched to points, then the result $A^*(\alpha_j^i)$, is homeomorphic to A' , where A' is constructed as A was, but using a different subset C' of the standard Cantor set. Finally, let $A(\alpha^i)$ be the union of the $A(\alpha_j^i)$, $j = 1, \dots, i$, and let $A^*(\alpha^i)$ be the union of the $A^*(\alpha_j^i)$, $j = 1, \dots, i$.

We will construct inductively a sequence of continuous maps $\{K_i\}$, $K_i: R^n \rightarrow R^n$, satisfying

- (1_i) $K_i = \text{identity on } f(S^{n-1})$ for $i \geq 1$,
- (2_i) $K_i = K_{i-1}$ on $f(A_{1/2} - A(\alpha^i))$,

- (3_i) $K_i f(\Sigma_i) = t_L$ for $t = d_i, \dots, d_i$,
 (4_i) $K_i f(\Sigma_i) \subset N_{1/i}(I_{jL})$ if $t \in I_j, j = 1, \dots, i$,
 (5_i) K_i is a homeomorphism on a neighborhood of $f(A_{1/2} - \bigcup_{j=1}^i \Sigma_{d_j})$.

Start the induction with $K_0 = \text{identity}$.

Assume by induction that K_{i-1} has been constructed. f imbeds $A(\alpha^i)$ and $K_{i-1}f$ imbeds $A^*(\alpha^i)$. We can apply (B) to each $A^*(\alpha_j^i)$ with $\varepsilon = 1/i$ and $f = K_{i-1}f$ to obtain G_i , where $G_i K_{i-1}$ will satisfy (1_i), (2_i), (4_i) and (5_i).

Next we apply (C) to just the $A^*(\alpha_j^i)$ which contains d_i , letting the f of (C) be $G_i K_{i-1}f$ and $t = d_i$. We obtain a map H_i , and $K_i = H_i G_i K_{i-1}$ will clearly satisfy (1_i)—(5_i).

Let $K = \lim K_i$. If we choose the α_j^i 's so that $\lim_{i \rightarrow \infty} \alpha^i(t) = 1/2$ for all t , then K exists and is a homeomorphism outside $f(S^{n-1} \times 1/2)$ because in a neighborhood of any such point, $K_i = K_{i+1}$ for i large enough (see (2_i)). By (3_i), $Kf(\Sigma_i) = t_L$ for $t \in D$. Furthermore, K is continuous on $f(S^{n-1} \times 1/2) = \bigcup_i \Sigma_i$ by (4_i). Since D is dense, $Kf(\Sigma_i) = t_L$ for all $t \in [-1, 1]$.

(E) $f|S^{n-1}$ extends to an imbedding f' of B^n into R^n .

PROOF. Let

$$\begin{aligned} F: B^n - \left(J \times \frac{1}{2}\right) &\rightarrow (B^n \cap A_{1/2}) - \left(S^{n-1} \times \frac{1}{2}\right) \\ &= \left\{ (x, s) \in A \mid x \in C \text{ and } 0 \leq s < \frac{1}{2} \right\} \end{aligned}$$

be the obvious homeomorphism which expands radially away from $J \times 1/2$. Define f' by

$$f'(x) = \begin{cases} KfF(x) & \text{if } x \in B^n - J \times \frac{1}{2} \\ f(x) & \text{if } x \in J \times \frac{1}{2} \end{cases}.$$

Since $F = \text{identity}$ on S^{n-1} , f' is an extension of f by (1) of (D). f' is seen to be well defined using (3) of (D). It is then easily verified that f' is an imbedding.

4. *Proof of Theorem 3.* If we let $2\tilde{B}^n$ be $2B^n$ with $2B^n \cap X_n$ pinched to 0, then we can assume that f extends to an imbedding of $2\tilde{B}^n$, since $f|B^{n-1} = 0$ is flat. Now it is sufficient to find an embedding $g: \Sigma^{n-1} \rightarrow \Sigma^n$, flat except at 0 and ∞ , which agrees with f on some neighborhood of 0, say rB^{n-1} , $r > 0$. For there are homeomorphisms $h_1, h_2: \Sigma^n \rightarrow \Sigma^n$, h_1 taking $g(B^{n-1})$ to $g(rB^{n-1}) = f(rB^{n-1})$ and h_2 taking $f(rB^{n-1})$ to $f(B^{n-1})$. Then $f(B^{n-1}) \subset h_2 h_1 g(\Sigma^{n-1})$ and f may be extended as required.

Without loss of generality we may assume that $f(\partial B^{n-1}) = f(S^{n-2}) = S^{n-2}$.

For there is an arc $A \subset f(B^{n-1})$ (the image of a radius) from $f(0)$ to a point $b \in f(\partial B^{n-1})$, which is locally flat mod $f(0)$, hence flat. Then there exists a map $s: \Sigma^n \rightarrow \Sigma^n$ which is the identity on $f(\partial B^{n-1})$, a homeomorphism off A and shrinks A to b . Then $sf(B^{n-1})$ is an $(n-1)$ -ball with boundary $f(\partial B^{n-1})$ which is locally flat except at the point b in its boundary. By [11] this ball is flat. Therefore $f(\partial B^{n-1})$ is unknotted in Σ^n , so a space homeomorphism takes it onto S^{n-2} . Furthermore we can assume $f(0) = 0$.

Let $j: \Sigma^{n-1} \rightarrow 2\tilde{B}^n - \partial B^{n-1}$ be a map satisfying

- (i) $j = \text{identity}$ in a neighborhood of 0 ,
- (ii) $j(\infty) = 0$,
- (iii) $j|(\Sigma^{n-1} - \{0, \infty\})$ is an imbedding, and

(iv) if ρ is a compactified ray in Σ^{n-1} beginning at 0 , then $j(\rho)$ lies in the plane P determined by X_n and ρ and winds once around ∂B^{n-1} as in Figure 2. In particular we require that $j(\rho)$ "represents" a generator of

$$H_1(2\tilde{B}^n - \partial B^{n-1}) = \mathbb{Z}.$$

To ensure local flatness later on, we assume that j is extended to a product neighborhood N of Σ^{n-1} , pinched at 0 and ∞ , such that $j|N - \{0, \infty\}$ is an imbedding into $2\tilde{B}^n - \partial B^{n-1}$. Now we wish to "unwind" $fj(N)$, rotating

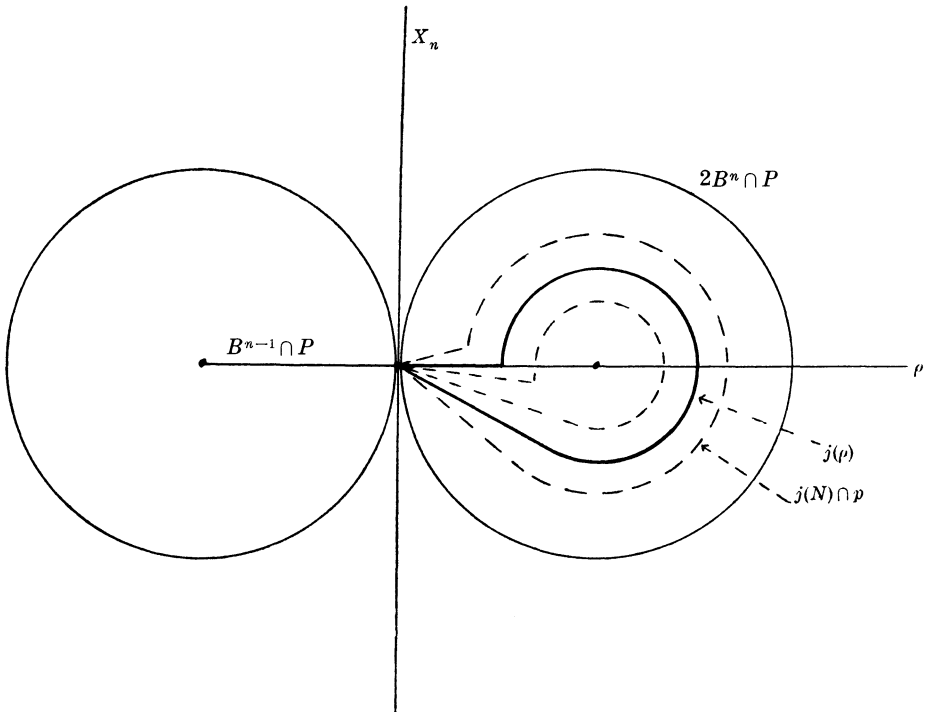


Figure 2

around S^{n-2} , so that $fj(0)$ and $fj(\infty)$ no longer coincide; then Σ^{n-1} will be flatly imbedded except at 0 and ∞ and will contain a neighborhood of $f(0)$ in $f(B^{n-1})$.

There is a natural homeomorphism of $\text{int } B^{n-1} \times S^1$ onto $\Sigma^n - S^{n-2}$ taking $0 \times S^1$ onto the compactified X_n -axis and $\text{int } B^{n-1} \times 0$ onto $\text{int } B^{n-1}$ (here S^1 is $[0, 2\pi]$ with $0 = 2\pi$). Identifying $\Sigma^n - S^{n-2}$ with $\text{int } B^{n-1} \times S^1$ in this way, let $p_1: \Sigma^n - S^{n-2} \rightarrow \text{int } B^{n-1}$ and $p_2: \Sigma^n - S^{n-2} \rightarrow S^1$ be the projections. Let $q: R^1 \rightarrow S^1$ be the universal covering space (where $q(x) \equiv x \pmod{2\pi}$). Since N is simply connected, the map $p_2 fj: N \rightarrow S^1$ lifts to a unique map $\lambda: N \rightarrow R^1$ satisfying $q\lambda = p_2 fj$ and $\lambda(0) = 0$. In effect, λ assigns to each point in N a "winding number" around S^{n-2} .

We now show that $\lambda(\infty) \neq 0$. Let ρ be a compactified ray in Σ^{n-1} as above, so ρ is an arc with end points 0 and ∞ . Recalling that $f(\partial B^{n-1}) = S^{n-2}$, $fj(\rho)$ represents a generator of $H_1(f(2\tilde{B}^n) - S^{n-2})$, since $j(\rho)$ represents a generator of $H_1(2\tilde{B}^n - S^{n-2})$. By excision on the pair $(\Sigma^n, f(2\tilde{B}^n))$, we see that $H_i(\Sigma^n - S^{n-2}, f(2\tilde{B}^n) - S^{n-2}) = 0$, ($i = 1, 2$), and hence the inclusion $f(2\tilde{B}^n) - S^{n-2} \subset \Sigma^n - S^{n-2}$ induces an isomorphism of first homology groups. Thus $fj(\rho)$ also represents a generator of $H_1(\Sigma^n - S^{n-2})$. Finally, $p_2: \Sigma^n - S^{n-2} \rightarrow S^1$ is a homotopy equivalence, so $p_2 fj(\rho)$ represents a generator of $H_1(S^1)$. Thus $\lambda(\infty) = \pm 2\pi$.

To unwind the map fj , let $\alpha: R^1 \rightarrow (-\pi, \pi)$ be a homeomorphism which is the identity on $(-\pi/2, \pi/2)$, and define $G: N \rightarrow \Sigma^n - S^{n-2} = \text{int } B^{n-1} \times S^1$ by $G(z) = (p_1 fj(z), q\alpha\lambda(z))$. The verification that G is an imbedding is routine, noting that $\lambda(0) \neq \lambda(\infty)$. If $z \in B^{n-1}$ is sufficiently close to 0, then $p_1 fj(z) = p_1 f(z)$ and $q\alpha\lambda(z) = q\lambda(z) = p_2 fj(z) = p_2 f(z)$. Thus $g = G|_{\Sigma^{n-1}}$ agrees with f near 0 and is locally flat off 0, $\infty \in \Sigma^{n-1}$, completing the proof.

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